

Mathematics Extension 1

HSC Marking Feedback 2019

Question 11

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Part (a)

Students should:

- use knowledge of gradients of straight lines to find the angle between two lines.

In better responses, students were able to find the:

- gradients of the given lines and used the formula $\theta = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ correctly to find the acute angle between the lines
- angle that each line makes with the horizontal and, by subtraction, found the correct angle between the lines.

Areas for students to improve include:

- making the variable y the subject so that the gradient can be read off from the equation
- ensuring the formula $\theta = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ is correctly written and used with particular attention to the signs in the numerator and denominator
- understanding the purpose and correct use of the absolute value
- understanding finding the point of intersection of the two lines has no bearing on the answer.

Part (b)

Students should:

- be able to solve an inequality with a variable in the denominator.

In better responses, students were able to:

- use the method of multiplying by the square of the denominator
- find the critical points and tested points on either side to reach the correct solution
- consider options to satisfy the given condition: $x + 1 > 0$ or $x + 1 < 0$
- use the graph of the hyperbola and the straight line $y = 2$ and the point of intersection to solve the inequality.

Areas for students to improve include:

- understanding the sign implications if there is a variable in the denominator of an

inequality

- working on algebraic skills when attempting to multiply by the square of the denominator
- practising solving quadratic inequalities
- avoiding writing that $-1 < x < -2$ on the number line when testing points or solving the quadratic inequality.

Part (c)

Students should:

- demonstrate their knowledge of circle geometry by using appropriate theorems to show that two angles in the given diagram are equal.

In better responses, students were able to:

- present a proof that used appropriate theorems supported by reasons that were clear and concise
- provide proofs that were well structured and easy to follow
- use conventional symbols and abbreviations correctly.

Areas for students to improve include:

- exercising caution in labelling angles correctly
- supporting each claim in a proof with a valid reason that is expressed clearly and concisely
- developing the proof in a logical manner
- distinguishing between similar and congruent triangles.

Part (d)

Students should:

- use their knowledge of the remainder theorem to find the quotient when a given polynomial is divided by a linear expression

In better responses, students were able to:

- execute the long division of the polynomial to correctly find the quotient $Q(x)$
- establish that the quotient $Q(x)$ is a quadratic and then correctly found the coefficients of this quadratic either by equating coefficients or by inspection.

Areas for students to improve include:

- understanding the remainder theorem and how to apply it
- accurately performing long division of a polynomial
- recognising that in this question the quotient $Q(x)$ needs to be a quadratic.

Part (e)

Students should:

- use their knowledge of double angles to express $\sin^2 4x$ in terms of $\cos 8x$ to integrate the

given trigonometric expression.

In better responses, students were able to:

- start with the double angle formula and logically worked towards the reduced form that needed to be integrated
- perform the integration with the proper treatment of the constants involved in the integrand.

Areas for students to improve include:

- recognising the need to reduce the given trigonometric function so that it can be integrated
- using a method to express $\sin^2 4x$ in terms of $\cos 8x$ rather than relying on memorisation of a formula
- using the Reference Sheet to help check their integration of standard expressions
- paying attention to the constants associated with standard integrals.

Part (f) (i)

Students should:

- apply their knowledge of the theory of probability to determine the probability of the required favourable event.

In better responses, students were able to:

- recognise that binomial probability modelling was needed and used the appropriate values in the relevant binomial term.

Areas for students to improve include:

- recognising when a problem involves a binomial probability term
- determining the correct values for p, q and n .

Part (f) (ii)

Students should:

- demonstrate that they are able to identify the modelling of a binomial probability to calculate the probability of a desired event.

In better responses, students were able to:

- make effective use of the complementary event to determine the probability.

Areas for students to improve include:

- understanding what 'at least' means in the context of probability problems
- clearly articulating that $P(\text{at least 2 wins}) = 1 - P(0 \text{ wins}) - P(1 \text{ win})$
- understanding that it is more efficient to present the probability using sigma notation rather than writing out all the terms in expanded form
- paying particular attention to use the correct values for p, q and n , and for the coefficient $\binom{n}{r}$.

Question 12

Part (a)

Students should:

- recognise a related rates question and understand the processes required to solve it.

In better responses, students were able to:

- show the use of the chain rule
- show their result for $\frac{dB}{dA}$
- evaluate B correctly and substituted it correctly into the chain rule.

Areas for students to improve include:

- using the calculator correctly
- differentiating to find $\frac{dB}{dA}$.

Part (b) (i)

Students should:

- identify when to use the auxiliary angle/transformation method to solve trigonometric equations.

In better responses, students were able to:

- find the values for R and α (in exact form) efficiently.

Areas for students to improve include:

- applying the auxiliary angle/transformation method skillfully
- knowing and using exact radian values.

Part (b) (ii)

Students should:

- use the mark value of a question as a guide to the complexity of solution required.

In better responses, students were able to:

- realise that the particle would be at rest at the end points of its motion and that this equated to the amplitude of their expression from part (i)

Areas for students to improve include:

- understanding the features of Simple Harmonic Motion
- reading the question carefully in order to answer the question that is asked.

Part (b) (iii)

Students should:

- understand the interrelatedness of the parts of a question
- use their result from part (i) to obtain an expression for velocity and solve a trigonometric equation to find a time.

In better responses, students were able to:

- realise that maximum velocity would be the amplitude of their expression for velocity
- understand that speed would be a positive value
- deduce that a 4th quadrant result to their trigonometric equation would still yield a positive value for time.

Areas for students to improve include:

- understanding the difference between speed and velocity
- developing their ability to solve trigonometric equations and working with exact trigonometric values.

Part (c)

Students should:

- take time to plan their approach to a proof, considering the facts needed and how to most easily obtain them.

In better responses, students were able to:

- use the definition of a parabola to determine that $PS = PM$
- find the length of PM using the fact that it was a vertical line and avoided using the distance formula
- use the formula for the equation of the tangent from the Reference Sheet to find the coordinates of R
- determine the length of SR without the use of the distance formula
- show that $SR = PS$ and used this to clearly justify why $\angle SPR = \angle SRP$
- use other methods such as the reflective property of the parabola to achieve the required result.

Areas for students to improve include:

- enhancing algebraic skills to avoid complicating algebraic manipulations.

Part (d) (i)

Students should:

- use the mark value of a question as a guide to the complexity of solution required.

In better responses, students were able to:

- to quickly and efficiently obtain this result.

Areas for students to improve include:

- knowing the method and the requirements of the question.

Part (d) (ii)

Students should:

- use the given equation from part (i) in order to structure their response to this part.

In better responses, students were able to:

- connect the right times and temperatures and work carefully through finding A , k and then $\frac{dT}{dt}$
- use exact values or calculator memories to maintain accuracy as they move through the stages of finding A , k and then $\frac{dT}{dt}$.

Areas for students to improve include:

- reading the question accurately to determine the relationships between variables
- using the calculator accurately
- showing all working out.

Question 13

Part (a)

Students should:

- apply their knowledge of calculus, algebra and arithmetic to evaluate a definite integral

In better responses, students were able to:

- differentiate $u = \cos^2 x$ correctly
- substitute correctly for u
- change the bounding limits and place them in the correct positions
- integrate and substitute correctly to arrive at the solution.

Areas for students to improve include:

- avoiding common errors such as
- providing the correct integrand,
- using the original limits
- providing the correct integrand but placing the limits incorrectly
- differentiating incorrectly
- making errors with negative signs

- avoiding careless substitution errors.

Part (b)

Students should:

- use the binomial theorem to find consecutive coefficients of $(5x + 2)^{20}$ in terms of k , equate them and solve for k using the expansion of $\binom{n}{k}$ to arrive at the solution.

In better responses, students were able to:

- equate the coefficients in terms of k and $k + 1$, expand and solve the resulting equation
- manipulate the expansion of the binomial coefficient and the indices successfully
- use guess, check and refine to arrive at the solution, provide the substitution line and supporting evidence.

Areas for students to improve include:

- using brackets carefully when writing out an expansion
- correctly recalling and using of the formula for $\binom{n}{k}$
- accurately applying algebraic skills when manipulating indices.

Part (c) (i)

Students should:

- make use of $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ to integrate and find v^2
- show specific use of given conditions to find the constant of integration.

In better responses, students were able to:

- manipulate the fractional indices and integrate successfully
- include and find the constant of integration
- show specific use of given conditions to order to arrive at the correct positive solution.

Areas for students to improve include:

- practising integration involving exponents with fractional indices
- ensuring the inclusion of the constant of integration
- including all reasoning in a 'show' question.

Part (c) (ii)

Students should:

- use an integration method to find t as a function of x and include the given conditions to find the constant of integration to arrive at an expression for x as a function of t .
- manipulate exponential expressions with fractional indices efficiently and reformat into logarithmic format.

In better responses, students were able to:

- use the reciprocal of $v = \frac{dx}{dt}$ to integrate to find an expression for t , use the given conditions to find the constant of integration
- use separation of variables and then integrated to find an expression for t and use the given conditions to find the constant of integration
- manipulate exponential equations into logarithmic format in order to provide an expression of x as a function of t .

Areas for students to improve include:

- fulfilling the requirements of the question
- providing an expression of x as a function of t
- practising the method of finding the reciprocal of $v = \frac{dx}{dt}$ to integrate to find an expression for t
- remembering to include the constant of integration.

Part (d) (i)

Students should:

- apply their knowledge of algebra, trigonometry and parameters to solve for t and x
- use the equations of motion, and the right-angled triangle to calculate a distance on an inclined plane.

In better responses, students were able to:

- solve for t and hence x , and use either Pythagoras' theorem or trigonometry to find the required distance
- find the Cartesian equation of motion and solved for x , and used either Pythagoras' theorem or trigonometry to find the required distance.

Areas for students to improve include:

- interpreting the requirements of the question correctly in order to find either x or t according to the given conditions.
- working carefully through multiple steps to arrive at the solution without making errors.

Part (d) (ii)

Students should:

- find the angle between a sloping path and a projectile at the point of impact
- use the components of velocity to solve $\tan \alpha = \left[\frac{\dot{y}}{\dot{x}} \right]$ to find the angle of impact and then calculating the angle with the plane
- use calculus techniques and the formula $\tan \theta = \left[\frac{m_1 - m_2}{1 - m_1 \times m_2} \right]$ to find the angle between the tangent at the point of impact and the equation of the sloping plane.

In better responses, students were able to:

- substitute the correct value of x from part (i) into the Cartesian equation to get the gradient of the tangent at A
- use the respective gradients in the formula $\tan \theta = \left[\frac{m_1 - m_2}{1 - m_1 \times m_2} \right]$ to arrive at the angle
- use the components of velocity to solve $\tan \alpha = \left[\frac{\dot{y}}{\dot{x}} \right]$ to obtain the angle of impact and then calculated the angle with the plane by appropriate subtraction.

Areas for students to improve include:

- avoiding errors when substituting values into the Cartesian equation, when substituting into $\tan \alpha = \left[\frac{\dot{y}}{\dot{x}} \right]$, or when substituting into $\tan \theta = \left[\frac{m_1 - m_2}{1 - m_1 \times m_2} \right]$.

Question 14

Part (a)

Students should:

- be able to perform proof by mathematical induction.

In better responses, students were able to:

- clearly show the inductive step using the correct assumption in the $n = k + 1$ case
- show their knowledge of algebra using factorial terms
- display sufficient working to indicate a correct proof and obtain full marks.

Areas for students to improve include:

- understanding how to manipulate expressions that involve factorials,
- recognising that $(k + 1)! = (k + 1)k!$.

Part (b) (i)

Students should:

- interpret the connection between the point of intersection of the two graphs and the root of the cubic equation.

In better responses, students were able to:

- equate functions to find points of intersection
- understand the connection between the points of intersection of two graphs, the roots of an equation and the zeros of a function.

Areas for students to improve include:

- interpreting the wording of a question
- understanding and obeying the instruction 'By considering graphs, explain...'
- giving sufficient information in explanations.

Part (b) (ii)

Students should:

- apply Newton's method correctly to find an approximation to the root of a polynomial.

In better responses, students were able to:

- use the Reference Sheet and Newton's method to fully show the required result.

Areas for students to improve include:

- developing accurate and efficient differentiation and algebraic skills.

Part (b) (iii)

Students should:

- be able to perform the algebraic manipulation to show $f(x_2) > 0$, after showing that $f(x_1) < 0$
- apply the theory connecting the zero of a continuous function to the sign either side of it.

In better responses, students were able to:

- realise that showing that the function values $f(x_1) < 0$ and $f(x_2) > 0$ were integral to achieving the required result
- use algebra and inequality arguments to show that $f(x_2) > 0$
- use a graphical approach arguing their solution from the shape of the cubic polynomial and the position of the stationary points and zeroes.

Areas for students to improve include:

- knowing how to manipulate inequalities
- solving, expanding and factorising algebraic expressions involving multiple terms and cubics.

Part (c) (i)

Students should:

- be able to differentiate trigonometric functions
- understand the relationship between the gradient at a point and common tangents.

In better responses, students were able to:

- use the connection between common tangents, gradients and equal derivative values.

Areas for students to improve include:

- understanding the difference between a function and a function value.

Part (c) (ii)

Students should:

- use the question numbering system to connect question parts when solving extended

problems

- use an appropriate trigonometric identity to achieve the required solution.

In better responses, students were able to:

- use an appropriate identity or concavity argument to show the required result
- understand the boundary conditions on the variables k and α and the connection between x_0 and $x_0 - \alpha$.

Areas for students to improve include:

- understanding the evenness or oddness of sine and cosine functions
- knowing the general results of trigonometric equations.

Part (c) (iii)

Students should:

- recognise the relationship between this part and part (ii) of the question
- be able to manipulate expressions involving trigonometric functions.

In better responses, students were able to:

- use part (ii) to forge a correct expression for k
- interpret the question to find k in terms of α only and realise that $x_0 = \alpha - x_0$.

Areas for students to improve include:

- developing an understanding of how the various parts of a question are related
- balancing time throughout the exam to allow more time to focus on difficult last questions.